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Low-profile skylight device

Abstract

The present invention relates to a low-profile skylight device. A frame of the device receives a window, wherein the frame that extends no more than 0.25 inch above a roof structure. The frame is comprised of a flange that allows the frame to attach to a roof structure using at least one fastener placed through at least one opening of the flange and/or at least one adhesive on a bottom surface of the flange. The membrane is preferably comprised of a waterproof, single-ply membrane that can be attached around the frame and flange to the roof surface via an adhesive bottom surface.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION (1) The present application claims priority to, and the benefit of, U.S. Provisional Application No. 63/422,587, which was filed on Nov. 4, 2022, and is incorporated herein by reference in its entirety.

FIELD OF THE INVENTION

(1) The present invention relates generally to the field of skylights. More specifically, the present invention relates to a low-profile skylight device. A frame of the device receives a window, wherein the frame extends no more than 0.25 inch above a roof structure. The frame is comprised of a flange that allows the frame to attach to a roof structure using at least one fastener placed through at least one opening of the flange and/or at least one adhesive on a bottom surface of the flange. The membrane is preferably comprised of a waterproof, single-ply membrane that can be attached around the frame and flange to the roof surface via an adhesive bottom surface. Accordingly, the present disclosure makes specific reference thereto. Nonetheless, it is to be appreciated that aspects of the present invention are also equally applicable to other like applications, devices, and methods

of manufacture.

BACKGROUND

(2) Installing skylights can be a difficult and frustrating process. This is especially true because skylights pose a tripping hazard on a roof. This is due to skylights being installed with curbing that extends several inches above a roof surface. In addition, installing existing skylights known in the art is a time-intensive and difficult process that typically requires special training.

(3) Therefore, there exists a long-felt need in the art for an improved skylight. There also exists a long-felt need in the art for a low-profile skylight device. More specifically, there exists a long-felt need in the art for a low-profile skylight device that prevents less of a tripping hazard than existing skylights known in the art. In addition, there exists a long-felt need in the art for a low-profile skylight device that can be easily installed. Further, there exists a long-felt need in the art for a low-profile skylight device that can be quickly installed without special/substantial training. Finally, there exists a long-felt need in the art for a low-profile skylight device that can be installed without curbing.

(4) The subject matter disclosed and claimed herein, in one embodiment thereof, comprises a low-profile skylight primarily comprised of a frame comprised of at least one flange, at least one window, and at least one membrane. The frame is preferably comprised of a frame that extends no more than 0.25 inch above a roof structure. The frame receives the window, which is preferably made from a transparent safety glass. In addition, the flange allows the frame to attach to a roof structure using at least one fastener placed through at least one opening of the flange and/or at least one adhesive on a bottom surface of the flange. The membrane is preferably comprised of a waterproof, single-ply membrane that can be attached around the frame and flange to the roof surface via an adhesive bottom surface.

(5) In this manner, the low-profile skylight device of the present invention accomplishes all the foregoing objectives and provides a low-profile skylight that prevents less of a tripping hazard than existing skylights known in the art. In addition, the device can be easily installed without special/substantial training. Finally, the device can be installed without curbing.

SUMMARY

(6) The following presents a simplified summary to provide a basic understanding of some aspects of the disclosed innovation. This summary is not an extensive overview, and it is not intended to identify key/critical elements or to delineate the scope thereof. Its sole purpose is to present some general concepts in a simplified form as a prelude to the more detailed description that is presented later.

(7) The subject matter disclosed and claimed herein, in one embodiment thereof, comprises a low-profile skylight. The device is primarily comprised of a frame comprised of at least one flange, at least one window, and at least one membrane. The frame of the device is preferably square or rectangular in shape. The frame is also preferably made from a corrosion-resistant metal material. The frame may be made of one continuous frame or a plurality of frame members that are attached together. The frame is preferably less than 0.25 inch in height to reduce the possibility of the device becoming a tripping hazard during roof construction/installation.

(8) At least one window is positioned within the frame. In one embodiment the window is received by at least one waterproof gasket of the frame that prevents water from leaking through the space between the frame and window when installed in a structure. The window may be made from a plurality of material types. In one embodiment, the window is made from a transparent safety glass.

(9) The frame can be attached to a roof structure during use using at least one flange. The flange may be comprised of at least one opening that allows at least one fastener to be inserted through the flange and secured to the structure. The fastener may be any type such as, but not limited to, a screw, a bolt, etc.

(10) In one embodiment, the bottom surface of the flange may additionally or alternatively be comprised of at least one adhesive. The adhesive allows the flange to further attach to a roof

structure. One embodiment of the flange may not have any openings and may only have the adhesive.

(11) The frame is further secured to a roof structure via at least one membrane. The membrane is preferably a single-ply membrane that is waterproof to prevent water from penetrating the frame. The bottom surface of the membrane may be comprised of at least one adhesive that allows the membrane to further attach to a roof structure.

(12) The device is also comprised of a method of using the device. First, a low-profile skylight device is provided comprising a frame, a window, and a membrane. Next, the frame is secured to a roof structure by placing at least one fastener through at least one opening in the flange and/or via at least one adhesive on a bottom surface of the flange. After the frame has been secured, at least one membrane can be placed around the frame and attached to the roof surface to create a waterproof seal around the frame.

(13) Accordingly, the low-profile skylight device of the present invention is particularly advantageous as it provides a low-profile skylight that prevents less of a tripping hazard than existing skylights known in the art. In addition, the device can be easily installed without special/substantial training. Finally, the device can be installed without curbing. In this manner, the low-profile skylight device overcomes the limitations of existing skylights known in the art.

(14) To the accomplishment of the foregoing and related ends, certain illustrative aspects of the disclosed innovation are described herein in connection with the following description and the annexed drawings. These aspects are indicative, however, of but a few of the various ways in which the principles disclosed herein can be employed and are intended to include all such aspects and their equivalents. Other advantages and novel features will become apparent from the following detailed description when considered in conjunction with the drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The description refers to provided drawings in which similar reference characters refer to similar parts throughout the different views, and in which:

(2) FIG. 1 illustrates a top view of one potential embodiment of a low-profile skylight device of the present invention in accordance with the disclosed architecture;

(3) FIG. 2 illustrates a bottom view of one potential embodiment of a low-profile skylight device of the present invention in accordance with the disclosed architecture;

(4) FIG. 3 illustrates a perspective view of one potential embodiment of a low-profile skylight device of the present invention while attached to a roof structure in accordance with the disclosed architecture; and

(5) FIG. 4 illustrates a flowchart of a method of using one potential embodiment of a low-profile skylight device of the present invention in accordance with the disclosed architecture.

DETAILED DESCRIPTION

(6) The innovation is now described with reference to the drawings, wherein like reference numerals are used to refer to like elements throughout. In the following description, for purposes of explanation, numerous specific details are set forth to provide a thorough understanding thereof. It may be evident, however, that the innovation can be practiced without these specific details. In other instances, well-known structures and devices are shown in block diagram form to facilitate a description thereof. Various embodiments are discussed hereinafter. It should be noted that the figures are described only to facilitate the description of the embodiments. They are not intended as an exhaustive description of the invention and do not limit the scope of the invention. Additionally, an illustrated embodiment need not have all the aspects or advantages shown. Thus, in other embodiments, any of the features described herein from different embodiments may be combined.

(7) As noted above, there exists a long-felt need in the art for an improved skylight. There also exists a long-felt need in the art for a low-profile skylight device. More specifically, there exists a long-felt need in the art for a low-profile skylight device that prevents less of a tripping hazard than existing skylights known in the art. In addition, there exists a long-felt need in the art for a low-profile skylight device that can be easily installed. Further, there exists a long-felt need in the art for a low-profile skylight device that can be quickly installed without special/substantial training. Finally, there exists a long-felt need in the art for a low-profile skylight device that can be installed without curbing.

(8) The present invention, in one exemplary embodiment, is comprised of a low-profile skylight device. The device is primarily comprised of a frame, comprised of at least one flange, at least one window, and at least one membrane. The frame of the device is preferably square or rectangular in shape and is preferably made from a corrosion-resistant metal material. The frame may be made of one continuous frame or a plurality of frame members that are attached together. The frame is preferably less than 0.25 inch in height to reduce the possibility of the device becoming a tripping hazard during roof construction/installation.

(9) Furthermore, at least one window is positioned within the frame. In one embodiment the window is received by at least one waterproof gasket of the frame that prevents water from leaking through the space between the frame and window when installed in a structure. The window may be made from a plurality of material types. In one embodiment, the window is made from a transparent safety glass.

(10) The frame can be attached to a roof structure during use using at least one flange. The flange may be comprised of at least one opening that allows at least one fastener to be inserted through the flange and secured to the structure. The fastener may be any type such as, but not limited to, a screw, a bolt, etc. In one embodiment, the bottom surface of the flange may additionally or alternatively be comprised of at least one adhesive. The adhesive allows the flange to further attach to a roof structure. One embodiment of the flange may not have any openings and may only have the adhesive.

(11) The frame is further secured to a roof structure via at least one membrane. The membrane is preferably a single-ply membrane that is waterproof to prevent water from penetrating the frame. The bottom surface of the membrane may be comprised of at least one adhesive that allows the membrane to further attach to a roof structure.

(12) The device is also comprised of a method of using the device. First, a low-profile skylight device is provided comprising a frame, a window, and a membrane. Next, the frame is secured to a roof structure by placing at least one fastener through at least one opening in the flange and/or via at least one adhesive on a bottom surface of the flange. After the frame has been secured, at least one membrane can be placed around the frame and attached to the roof surface to create a waterproof seal around the frame.

(13) Accordingly, the low-profile skylight device of the present invention is particularly advantageous as it provides a low-profile skylight that prevents less of a tripping hazard than existing skylights known in the art. In addition, the device can be easily installed without special/substantial training. Finally, the device can be installed without curbing. In this manner, the low-profile skylight device overcomes the limitations of existing skylights known in the art.

(14) Referring initially to the drawings, FIG. 1 illustrates a top view of one potential embodiment of a low-profile skylight device **100** of the present invention in accordance with the disclosed architecture. The device **100** is primarily comprised of a frame **110**, comprised of at least one flange **130**, at least one window **150**, and at least one membrane **160**. The frame **110** of the device **100** is preferably square or rectangular in shape. However, the frame **110** may be any shape known in the art. The frame **110** is preferably made from a corrosion-resistant metal material such as, but not limited to, stainless steel or aluminum. The frame **110** may be any color known in the art to blend into a roofline. The frame **110** may be made of one continuous frame or a plurality of frame

members **120** that are attached together. The frame members **120** may be attached via welds or removable/fixed fasteners.

(15) At least one window **150** is positioned within the frame **110**. More specifically, in one embodiment the window **150** is received by at least one waterproof gasket **122** of the frame **120**. The gasket **122** prevents water from leaking through the space between the frame **110** and window **150** when installed in a structure.

(16) The window **150** may be made from a plurality of material types. In one embodiment, the window **150** is made from a transparent safety glass. However, the window **150** may be made from any material known in the art. In addition, one embodiment of the window **150** may be comprised of a tinted window to reduce sunlight transmission through the window **150**.

(17) The frame **110** can be attached to a roof structure **10** during use. More specifically, each frame member **120** may be comprised of at least one flange **130**, as seen in FIG. 2 and FIG. 3. The flange **130** is preferably made from the same material as the frame **110**. The flange **130** is also comprised of at least one opening **140** that allows at least one fastener **180** to be inserted through the flange **130** and secured to the structure **10**. The fastener **180** may be any type such as, but not limited to, a screw, a bolt, etc. In one embodiment, the flange **130** may be made from a flexible material that allows the flange **130** to be easily bent to the slope and/or shape of a roof structure **10**.

(18) In one embodiment, the bottom surface **142** of the flange **130** may additionally or alternatively be comprised of at least one adhesive **134**. The adhesive **134** allows the flange **130** to further attach to a roof structure **10**. One embodiment of the flange **130** may not have any openings **140** and may only have the adhesive **134**.

(19) The frame **110** is further secured to a roof structure **10** via at least one membrane **160**, as seen in FIG. 3. The membrane is preferably a single-ply membrane, but may be any membrane known in the art. The membrane **160** is preferably a waterproof membrane to prevent water from penetrating the frame **110**. The bottom surface **170** of the membrane **160** may be comprised of at least one adhesive **172**. The adhesive **172** allows the membrane **160** to further attach to a roof structure **10**. The adhesives **134,172** of the device **100** may be any roofing or other related adhesive known in the art. It should be noted that the membrane **160** allows roofing materials such as, but not limited to, shingles or metal roofing to be easily installed over the membrane **160**.

(20) The frame **110** is preferably less than 0.25 inch in height. This reduces the possibility of the device **100** becoming a tripping hazard during roof construction/installation. The frame **110** is preferably flush with the membrane **160** and tapers to match the roof surface **10**.

(21) FIG. 4 illustrates a flowchart of a method of using **200** one potential embodiment of a low-profile skylight device **100** of the present invention in accordance with the disclosed architecture. The device **100** is also comprised of a method of using **200** the device **100**. First, a low-profile skylight device **100** is provided, the device **100** is comprised of a frame **110**, a window **150**, and a membrane **160** [Step 202]. Next, the frame **110** is secured to a roof structure **10** by placing at least one fastener **180** through at least one opening **140** in the flange **130** and/or via at least one adhesive **132** on a bottom surface of the flange **130** [Step 204]. After the frame **110** has been secured, at least one membrane **160** can be placed around the frame **110** and attached to the roof surface **10** to create a waterproof seal around the frame **110** [Step 206].

(22) Certain terms are used throughout the following description and claims to refer to particular features or components. As one skilled in the art will appreciate, different persons may refer to the same feature or component by different names. This document does not intend to distinguish between components or features that differ in name but not structure or function. As used herein “low-profile skylight device” and “device” are interchangeable and refer to the low-profile skylight device **100** of the present invention.

(23) Notwithstanding the foregoing, the low-profile skylight device **100** of the present invention and its various components can be of any suitable size and configuration as is known in the art without affecting the overall concept of the invention, provided that they accomplish the above-

stated objectives. One of ordinary skill in the art will appreciate that the size, configuration, and material of the low-profile skylight device **100** as shown in the FIGS. are for illustrative purposes only, and that many other sizes and shapes of the low-profile skylight device **100** are well within the scope of the present disclosure. Although the dimensions of the low-profile skylight device **100** are important design parameters for user convenience, the low-profile skylight device **100** may be of any size, shape, and/or configuration that ensures optimal performance during use and/or that suits the user's needs and/or preferences.

(24) Various modifications and additions can be made to the exemplary embodiments discussed without departing from the scope of the present invention. While the embodiments described above refer to particular features, the scope of this invention also includes embodiments having different combinations of features and embodiments that do not include all the described features.

Accordingly, the scope of the present invention is intended to embrace all such alternatives, modifications, and variations as fall within the scope of the claims, together with all equivalents thereof.

(25) What has been described above includes examples of the claimed subject matter. It is, of course, not possible to describe every conceivable combination of components or methodologies for purposes of describing the claimed subject matter, but one of ordinary skill in the art may recognize that many further combinations and permutations of the claimed subject matter are possible. Accordingly, the claimed subject matter is intended to embrace all such alterations, modifications, and variations that fall within the spirit and scope of the appended claims.

Furthermore, to the extent that the term “includes” is used in either the detailed description or the claims, such term is intended to be inclusive in a manner similar to the term “comprising” as “comprising” is interpreted when employed as a transitional word in a claim.

Claims

1. A skylight device comprising: a frame comprised of a flange; a window; and a membrane; and wherein the frame is less than 0.25 inches in height, is flush with the membrane, and tapers to match a surface of a roof; wherein the flange is a flexible flange configured to bend to match the surface of the roof; and wherein the window is a tinted safety glass window.
 2. A skylight device of claim 1, wherein a first bottom surface of the flange is comprised of a first adhesive.
 3. A skylight device of claim 2, wherein a first bottom surface of the membrane is comprised of a second adhesive.
 4. A skylight device of claim 1, wherein the frame is comprised of a square frame.
 5. A skylight device of claim 1, wherein the frame is comprised of a corrosion-resistant metal material.
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